

# Inhibitory control mediates the association between perinatal PM<sub>2.5</sub> exposure and childhood obesity in children in the PROGRESS cohort, Mexico City

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## Title Page

**Title:** Inhibitory Control Mediates the Association Between Perinatal PM2.5 Exposure and Childhood Obesity in children in the PROGRESS Cohort, Mexico City

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### Abstract

**Background:** Perinatal exposure to particulate matter 2.5 (PM<sub>2.5</sub>) is linked to both neurotoxicity and childhood obesity. Given the role of inhibitory control in eating behavior, we propose that early-life PM<sub>2.5</sub> disrupts inhibitory control, contributing to obesity as part of a causal pathway.

**Methods:** In the PROGRESS cohort (n = 434) in Mexico City, we modeled ambient PM<sub>2.5</sub> exposure during pregnancy and the first year of life. Inhibitory control was assessed at age 4 using the Conners Kiddie Continuous Performance Test, and repeated adiposity measures were tracked from ages 4 to 8. We conducted latent profile analysis to identify profiles of inhibitory control and applied multinomial causal mediation analyses to estimate the natural total, direct, and indirect (i.e., mediated) effects of inhibitory control on the prenatal and first-year-of-life PM<sub>2.5</sub> exposure with childhood obesity-related outcomes, using separate models for each exposure window.

**Results:** Our analysis show a significant natural indirect effect between first-year postnatal PM<sub>2.5</sub> concentrations and greater body mass index (BMI) ( $\beta=1.86\text{kg/m}^2$ , 95%CI: 0.72, 3.19), BMI z-score ( $\beta=0.11$ , 95%CI: 0.01, 0.25), and percent body fat ( $\beta=3.04\%$ , 95%CI: 1.26, 5.04) at 8 years, and BMI change ( $\beta=0.43\text{kg/m}^2$ , 95%CI: 0.14, 0.80) from ages 4 to 8 years mediated through inhibitory control. No evidence of mediation by inhibitory control was found for prenatal PM<sub>2.5</sub> predicting obesity-related outcomes.

**Conclusions:** PM<sub>2.5</sub> exposure in the first year of life indirectly predicts childhood obesity through a pathway mediated by poorer inhibitory control. This new understanding of the underlying obesogenic mechanisms of air pollution can significantly inform policy and clinical interventions.

**Keywords:** *Air Pollution, Inhibitory Control, Childhood Obesity, Causal Mediation Analysis, Neurodevelopment, PM<sub>2.5</sub>*

**Plane Language Summary:** Perinatal exposure to particulate matter 2.5 (PM<sub>2.5</sub>) has been shown to have neurotoxic effects and can lead to obesity. This study examined whether changes in inhibitory control, an executive function previously associated with obesity, from prenatal and early postnatal PM<sub>2.5</sub> exposure mediate its association with obesity-related outcomes in school-aged children. High exposure early in infancy is linked to altered inhibitory control, which in turn increases the risk of obesity, likely through more impulsive eating behaviors. These findings suggest that reducing early exposure to PM<sub>2.5</sub> could support healthier cognitive and metabolic development in children.

## INTRODUCTION

Childhood overweight and obesity (OWOB) is an alarming epidemic and a vexing public health threat that affects nearly 15 million children across the United States [1]. This issue is not confined to the United States alone—it is a growing global problem, with rising rates of childhood OWOB posing severe health risks worldwide. Notably, there is an increasing prevalence of childhood OWOB in Mexico, with 1 in 5 school-age children suffering from the disorder [2]. Statistics suggest that the significant increases in the prevalence of childhood OWOB over the past decade rank Mexico first in Latin America, with estimates that ~36% of children aged 5 – 11 years have overweight or obesity [3, 4]. Childhood OWOB is a metabolic and inflammatory disorder and a risk factor for multiple comorbidities, including diabetes, heart disease, hypertension [5-7], emotional distress (e.g., psychological conditions) [8], and academic underperformance [9, 10]. Childhood OWOB tends to track into adult life, which will likely raise the risk of developing comorbidities at increasingly younger ages.

Typically, childhood OWOB is ascribed to behavioral and physical attributes such as inadequate lifestyle behaviors relating to diet, exercise, and genetic predisposition. However, accumulating evidence has demonstrated that environmental exposures can also cause obesity. The term “*obesogen*” has been coined to describe environmental exposures that may cause weight gain. These exposures include several organic chemicals, such as pesticides, but also airborne pollutants, such as particulate matter (PM) [11-13]. There has been a general assumption that these exposures impact metabolism, potentially altering how food is digested, how energy is expended, or how fat, protein, and carbohydrates are stored. However, another possibility—perhaps conceptually simpler—is that obesogenic exposures may instead alter eating behaviors, leading to a preference for higher-calorie foods, increased food consumption, or both. Such a pathway would represent a behavioral mechanism that has not yet been studied. We propose that air pollution may operate through neurobehavioral pathways that regulate inhibitory control—an executive function previously associated with obesity.

PM is a complex blend of organic and inorganic particles suspended in the air from automobile exhaust, industrial releases, non-anthropogenic sources (e.g., dust), and specifically, an active volcano in Mexico City. PM is classified based on its aerodynamic diameter. PM<sub>2.5</sub> refers to particles less than 2.5 µm in diameter, often referred to as fine particulate matter. PM<sub>2.5</sub> has been shown to have obesogenic properties that can disrupt the human body’s metabolic homeostasis [14, 15], contributing to energy imbalance [16], inducing systemic oxidative stress and inflammation [17, 18], and triggering insulin resistance [19], all of which are associated with weight gain. Timing of exposure matters, and PM<sub>2.5</sub> *in utero* exposure has been shown to program increases in children’s weight later in life [20]. Several studies of postpartum exposure also show that early-life PM<sub>2.5</sub> exposure is associated with elevated body mass index (BMI) [21-26].

While the literature identifies multidimensional mechanisms as drivers of childhood OWOB, emerging evidence highlights the role of eating-related neurobehaviors, specifically inhibitory control. Conceptually, poorer inhibitory control can lead to impulsive eating, which, over time, leads to obesity. Inhibitory control is a fundamental component of executive functions and refers to the ability to suppress inappropriate behavioral and emotional predispositions or responses that may be triggered extrinsically by focusing on more appropriate predispositions [27, 28]. It emerges in early childhood, developing rapidly between 3 and 6 years of age [29, 30]. As the developmental trajectory enters adolescence and early adulthood, poor inhibitory control has been established as a trait that can influence eating behavior [31]. Research has shown that ineffective inhibitory control is more prevalent in children who are overweight or obese compared to non-overweight or non-obese children [32, 33] and is directly linked to disinhibited eating behaviors [31, 34], theoretically leading to a significant increase in body fat. Most studies are, however, cross-sectional, and few have addressed upstream environmental factors that may

program poor inhibitory control. We propose that early-life exposure to  $PM_{2.5}$  is a common early-life neurotoxicant via neuroinflammation and oxidative stress that can program poorer inhibitory control, partially explaining the obesogenic properties of  $PM_{2.5}$ .

Although accumulating evidence supports separate links between (1)  $PM_{2.5}$  and childhood OWOB and (2) poor inhibitory control and childhood OWOB, to our knowledge, no study has yet addressed whether these findings are linked. In other words, does  $PM_{2.5}$  exposure increase impulsivity and eating behaviors in children that, over time, produce higher BMI and weight gain? If so, then inhibitory control may act as a mediator in the relationship between early-life  $PM_{2.5}$  exposure and childhood OWOB. We propose to address this question in a prospective pre-birth cohort with state-of-the-art prospective measures of prenatal and postnatal air pollution exposure, inhibitory control, and obesity over more than a decade of longitudinal follow-up. We identify a statistically significant natural indirect effect linking  $PM_{2.5}$  exposure during the first year of life to greater obesity-related outcomes in school-aged children, operating through a pathway mediated by poorer inhibitory control. These findings reinforce the importance of sustained efforts to inform child health and environmental policies, integrating both behavioral and environmental interventions aimed at reducing air pollution and protecting children's health and development.

## METHODS

### Study Population

The Programming Research in Obesity, Growth, Environment, and Social Stressors (PROGRESS) study recruited pregnant women who received prenatal care through the Mexican Social Security System (Instituto Mexicano del Seguro Social, IMSS) between July 2007 and February 2010. IMSS provides healthcare to affiliated private sector employees, the majority of whom are low- to middle-income workers and their families. The recruiting eligibility criteria included being a pregnant woman over 18 years of age, less than 20 weeks of gestation, who planned to stay in Mexico City for the next 3 years. Other criteria include access to a telephone, no major disease in medical history, and not using steroid or anti-epilepsy medications or consuming alcohol on a daily basis. The recruiting procedures were approved by institutional review boards at the Harvard School of Public Health, Icahn School of Medicine at Mount Sinai, and the Mexican National Institute of Public Health. Informed written consent was obtained from all maternal parents during the first visit, and all methods were performed in accordance with the relevant guidelines and regulations. A total of 948 mothers were enrolled in pregnancy and gave birth to a live child. Gestational age was based on the last menstrual period and standardized physical examination to determine gestational age at birth [35]. Participants were enrolled between 2007 and 2011, with most pregnancies occurring in 2007–2008. Births occurred from late 2007 through mid-2011, encompassing the entire cohort recruitment period. Since 2010, we have successfully followed 650 of these mother-children dyads. For this analysis, we used 434 full-term children with complete data on air pollution, inhibitory control (Kiddie Continuous Performance Test 2<sup>nd</sup> Edition™) at 4 years, covariates, and obesity measures from ages 4–10. Participants included in the final analytic sample were similar to the full available follow-up sample with respect to child sex ( $p = 0.939$ ), socioeconomic status ( $p = 0.558$ ), and maternal age at delivery ( $p = 0.227$ ). The study follows the approved protocols from the institutional review boards [IRB] of the Icahn School of Medicine at Mount Sinai [IRB protocol number: 12-00751A], and all three committees [Research, Ethics in Research, and Biosafety] were included in the IRB at the National Institute of Public Health in Cuernavaca, Mexico. Written informed consent was obtained from all participants at the initial visit, and all procedures were conducted in accordance with applicable guidelines and regulations.

### Body Mass Index

Children were followed-up at for adiposity assessments at two study visits: the first at mean 4.8 years (N = 514; range = 4.0–6.8) and the second at mean 9.7 years (N = 434; range = 8.1–11.7). Trained staff collected child weight and standing height twice at each study visit using a professional digital scale (Health-o-meter, McCook, IL) and averaged the values. BMI z-scores were estimated using the World Health Organization guidelines [36]. At visit 1 and 2, we estimated fat mass using Tetrapolar bioelectrical impedance with the InBody 230 (Biospace Co, Ltd, Cerritos, CA). We examined both BMI and BMI z-scores, providing a more comprehensive assessment of body size and growth. BMI offers an absolute measure of body mass relative to height, while BMI z-scores account for age- and sex-specific growth patterns, enabling better comparisons across developmental stages.

### **Prenatal and Postnatal PM<sub>2.5</sub> Prediction**

Daily ambient PM<sub>2.5</sub> concentrations were estimated using a hybrid satellite-based regression (LUR) hybrid PM<sub>2.5</sub> prediction model [37, 38] from birth through one year postpartum. Prenatal exposure was calculated as the average PM<sub>2.5</sub> concentration across the entire pregnancy, while daily estimates from the postnatal period were averaged to obtain a first-year exposure metric. Moderate-resolution imaging spectroradiometer (MODIS) satellite-derived aerosol optical depth (AOD) measurements were used to predict daily PM<sub>2.5</sub> concentration starting in 2004 across the greater Mexico City region. We employed ground PM<sub>2.5</sub> measurements from 12 monitoring stations maintained by the atmospheric monitoring system of the Mexico City government, which is publicly available. A multiyear geospatial database was constructed with satellite-measured AOD, PM monitor data, and all spatial and temporal predictors (such as land use, population data, meteorology, and atmospheric conditions). We then applied extreme gradient boosting (XGBoost) and inverse-distance weighting to yield residence-specific estimates of PM<sub>2.5</sub> exposure. Model performance was excellent with cross-validated mean absolute errors (MAE) of 3.68 and 9.20  $\mu\text{g}/\text{m}^3$ , respectively, compared to mean absolute deviations from the median (MAD) of 8.55 and 15.64  $\mu\text{g}/\text{m}^3$ . Predicted PM<sub>2.5</sub> at each participant's residence in relation to the 1x1 km grid locations showed sufficient variability to assess associations with health. Due to Mexico's climate, almost all participants had windows open 24 hours a day ( $\geq 94\%$ ). This removes the major concern of indoor–outdoor pollution ratio that commonly exists in studies conducted in developed countries. Although there are substantial sources of indoor air pollution, our estimates of ambient exposure levels may be more relevant for estimating the impact on health in Mexico compared to ambient exposure models in other developed countries.

### **Neurobehavioral Measurements**

We used the Conner's Kiddie Continuous Performance Test 2<sup>nd</sup> Edition™ (Conners K-CPT 2™) to assess neurobehavior in PROGRESS children at the 4 year visit. The K-CPT is a well-established tool constructed to assess attention deficits in children ages 4 to 7 [39]. K-CPT takes only 7.5 minutes to complete, making it appropriate for younger children. Results from the measure can be used for clinical assessment, early identification, and educational classification. The assessment is designed to assess inattentiveness, impulsivity, sustained attention, and vigilance by giving the subject a visual cue and an appropriate response. The K-CPT consists of five blocks, each containing 40 trials. Each block is further divided into two sets of 20 trials, alternating between short (1.5-second) and long (3.0-second) inter-stimulus intervals. During the assessment, children are seated facing a computer screen while images appear one at a time for 500 milliseconds. Children are instructed to respond by pressing the space bar or using a wired mouse only when a target image is shown. This measure has been validated in diverse populations, including Hispanic participants, and specifically within samples of Mexican children [40, 41]. The incorrect responses are recorded as errors such as omission error (OE; i.e., did not respond when instructed to) and commission error (CE; i.e., responded when instructed not to); higher scores indicate a greater rate of omission or commission errors. Perseverations were recorded as random, repetitive, or anticipatory responses made within 100 ms of the stimulus, with higher scores indicating a greater

frequency of perseverative errors. Hit reaction time (HRT; response speed) and the standard error of HRT (HRT-SE) are indexed by the time used from cue to responses and the variation throughout the test.

## Covariates

Previous literature recommends a number of covariates that are associated with obesity and neurobehavioral outcomes in children [42-44]. Information about the child's sex, maternal age at birth, socioeconomic status (SES), date of birth, and child age at visit was collected through direct interviews with the mothers at our research facilities at the Mexican National Institute of Perinatology. SES of participant families and 13 variables derived from prenatal questionnaire results were used to classify participants into categories based on the SES index created by the Asociación Mexicana de Agencias de Investigación de Mercados y Opinión Pública (AMAI) [45]. Our study classified participants as low, medium, and high SES on a relative scale within our cohort. Women ( $n = 7$ ) who reported smoking at enrollment were classified as prenatal smokers and were excluded from the analysis due to low observations. Covariates included in the final model were child sex, maternal age at birth, child age at neurocognitive test, and relative SES.

## Statistics and Reproducibility

### *K-CPT-based Latent Profile Analysis*

Because the K-CPT assesses multiple inhibitory control outcomes to explain a complex neurobehavior phenotype [46], we used latent profile analysis (LPA) to reduce the data into discrete categories. Specifically, we used LPA on the five K-CPT tasks (CE, OE, HRT, HRT-SE, and Perseverations) to objectively classify children into discrete data-driven inhibitory control (i.e., neurobehavior) profile scores using a model-based clustering technique. This technique was a normal mixture model fitted via an expectation-maximization algorithm in the “mclust” package in R [version 4.3; 47]. LPA assumes that the population consists of a number of unobserved subgroups that are referred to as latent profiles. Children within a given latent profile will show similarities in their K-CPT scores. For example, children in the same latent profile may demonstrate high scores on tests for CE and HRT-SE and low scores on OE, suggesting the child is continuously responding to cues without carefully assessing whether they should respond or not. Sharing a latent profile does not require consistent high or low scoring across all scores used in deriving the latent class. Instead, it identifies groups of subjects who have a similar performance pattern across the five K-CPT measures. We used the Bayesian information criteria (BIC) to select the number of latent profiles in our final finite mixture model [48]. Specifically, the lowest BIC value across models assuming different numbers of latent profiles indicates the best fit and minimizes cross-classification probabilities.

### *Mediation Analysis*

To assess the assumptions underlying causal mediation, we considered the conditions necessary to identify natural direct and indirect effects: no unmeasured confounding between exposure and outcome, exposure and mediator, or mediator and outcome after accounting for measured covariates. These conditions were addressed by maintaining the temporal order of data collection, with PM<sub>2.5</sub> exposure assessed before inhibitory control and adiposity outcomes measured several years later, and by adjusting all models for maternal age, socioeconomic status, and child sex, which are established predictors of both exposure and outcome in this cohort. Although these assumptions cannot be directly verified, they are considered plausible, and any remaining unmeasured confounding is acknowledged as a potential limitation.

We employed a causal mediation analysis framework to decompose the total effect of PM<sub>2.5</sub>

exposure (as a continuous linear term) on childhood OWOB outcomes into natural direct effects (NDE) and natural indirect effects (NIE); the indirect effect represents the effect mediated through inhibitory control changes, and the direct effect represents the effect on an outcomes through pathways other than inhibitory control, as characterized by the latent profiles. This approach allows us to assess the extent to which neurobehavioral changes contribute to the risk of obesity independently of the impact of air pollution, operating through pathways.

We conducted a mediation analysis using the counterfactual framework, which involves estimating the potential outcomes under different scenarios of mediator (in this case inhibitory control profiles) and exposure (PM<sub>2.5</sub>) levels. Let Y denote a continuous outcome. Let A denote the exposure, M denotes a categorical mediator with k levels, and X denotes a vector of covariates potential observed confounders, for a change in the exposure from level  $\alpha^*$  to level  $\alpha$ , the natural indirect and direct effects can be denoted non-parametrically as:

$$\begin{aligned} NIE &= E(Y_{\alpha}M_{\alpha} - Y_{\alpha}M_{\alpha^*}|x) = \sum_m E(Y|A = \alpha, M = m, X = x) \{p(M = m|A = \alpha, X = x) - p(M = m|A = \alpha^*, X = x)\} \\ NDE &= E(Y_{\alpha}M_{\alpha^*} - Y_{\alpha^*}M_{\alpha^*}|x) \\ &= \sum_m \{E(Y|A = \alpha, M = m, X = x) - E(Y|A = \alpha^*, M = m, X = x)\} p(M = m|A = \alpha^*, X = x) \end{aligned}$$

The NIE compares the average difference in the counterfactual outcome when fixing the exposure level to  $\alpha$ , while the mediator varies from the level it would have taken if the exposure was set to  $\alpha^*$  as compared to  $\alpha^*$ . The NDE compares the average difference in the counterfactual outcomes for a change in exposure level from  $\alpha^*$  to  $\alpha$  while fixing the mediator to the level it would have taken if the exposure is set to  $\alpha^*$ .

We utilized linear regression models for continuous outcomes, adjusting for covariates such as maternal age, socioeconomic status, and child's sex, and used multinomial logistic regression to model the categorical latent IC phenotypes as a mediator. The significance of the mediation effect was assessed using bootstrap methods to generate 95% confidence intervals (CI) for the indirect effect. We used nonparametric bootstrapping with 2,000 iterations, resampling the original dataset with replacement at each iteration to estimate the total, direct, and indirect effects. In each bootstrap sample, regression models were re-estimated and effect estimates recalculated accordingly. This approach follows the counterfactual framework proposed by Valeri and VanderWeele [49], which accommodates categorical mediators and interactions between the mediator and exposure in the outcome model. All analyses were conducted in R. We also conducted an analysis to assess the effect of prenatal PM<sub>2.5</sub> exposure but found no evidence of mediation. As a result, these findings were excluded from the final analysis for clarity.

## Results

### *Sample Characteristics*

Table 1 outlines a comprehensive overview of the socio-demographic characteristics of the study participants. It also includes detailed data on PM<sub>2.5</sub> concentrations and summarizes BMI outcome measures. The study consists of 434 children, showing a near-equal distribution between males (n = 219) and females (n = 215), with the average age at the second outcome visit being almost 10 years. The majority of participants (52.5 %) hailed from lower socioeconomic backgrounds. The mean  $\pm$  SD for prenatal and postnatal PM<sub>2.5</sub> concentrations were 22.7  $\mu\text{g}/\text{m}^3 \pm 2.6$  and 22.7  $\mu\text{g}/\text{m}^3 \pm 2.5$ , respectively. The mean  $\pm$  SD for BMI, BMI z-score, and percent body fat at age 8, and BMI change between ages 4 and 8 were 18.7  $\text{kg}/\text{m}^2 \pm 3.6$ , 0.8  $\pm 1.3$ , 30.0%  $\pm 8.8$ , and 3.1  $\text{kg}/\text{m}^2 \pm 2.7$ . Approximately 42.9% of the children were classified as having a high BMI. Furthermore, Table 1 provides the average performance on each K-CPT measure. Participants retained in the analytic sample did not differ significantly from

those lost to follow-up in baseline BMI, socioeconomic status, child sex, or age at assessment (all  $p > 0.05$ ).

### *Latent Profiles of K-CPT*

The latent profile analysis (LPA) identified distinct patterns of underlying latent inhibitory control profiles among children across five specific performance indicators from the K-CPT, including 1) CE, 2) OE, 3) HRT, 4) HRT-SE, and 5) Perseverations. The varying volume, varying shape, and varying orientation (VVO) model, which assigns a unique covariance matrix to each component, identified three latent groups based on children's neurobehavioral profiles, emerging as the optimal solution, characterized by the lowest BIC value. The means of each performance indicator for the three latent groups are as follows (Figure 1):

**Group 1 ( $n = 70$ )** is characterized by moderate scores across most indicators but relatively higher omission and hit standard error scores, with an average Commissions Error T-score of 49.4, an average Omission Error T-score of 58.7, an average Perseverations T-score of 54.9, an average Hit Reaction Time T-score of 54.9, and an average Hit Standard Error T-score of 56.9. This suggests a pattern of inattentiveness combined with variability in reaction times. **Group 2 ( $n = 178$ )** shows generally lower scores on all indicators, indicating a profile with fewer errors and more consistent performance, possibly representing a group with better attention and fewer impulsivity issues. This group has an average Commissions Error T-score of 51.4, an average Omission Error T-score of 46.3, an average Perseverations T-score of 46.5, an average Hit Reaction Time T-score of 46.7, and an average Hit Standard Error T-score of 46.9. **Group 3 ( $n = 186$ )** is marked by notably higher scores in commissions and perseverations, pointing to a profile of high impulsivity and repetitive responding, along with moderate performance in other indicators. The average Commissions Error T-score is 64.4, the average Omission Error T-score is 46.6, the average Perseverations T-score is 72.8, the average Hit Reaction Time T-score is 48.8, and the average Hit Standard Error T-score is 55.8. These latent groups provide a nuanced understanding of the neurobehavioral patterns observed in children and highlight the variability in their responses to the K-CPT.

### *Results of Mediation Analysis*

The mediation analysis assessed the total, direct and indirect effects of first-year postnatal  $PM_{2.5}$  exposure on various obesity-related outcomes, with inhibitory control profiles serving as the mediator (see Figure 2). The following outcomes were evaluated: BMI, BMI z-score, and percent body fat at age 8, and BMI change between ages 4 and 8.

The total effect estimates associated with a  $10 \mu g/m^3$  increase in first-year  $PM_{2.5}$  exposure were as follows:  $0.1 \text{ kg/m}^2$  for BMI (95% CI:  $-1.2, 1.4$ ),  $-0.01$  for BMI z-score (95% CI:  $-0.5, 0.5$ ),  $0.4\%$  for percent body fat (95% CI:  $-2.9, 3.6$ ), and  $0.3 \text{ kg/m}^2$  for BMI change (95% CI:  $-0.7, 1.2$ ). Exposure to  $PM_{2.5}$  in the first year postnatally was associated with both direct and indirect effects on BMI, BMI z-score, percent body fat, and BMI change. The direct effect estimates per  $10 \mu g/m^3$  for BMI was  $-0.9 \text{ kg/m}^2$  (95% CI:  $-3.8, 1.4$ ), while the indirect effect, mediated through behavioral changes, was  $1.9 \text{ kg/m}^2$  (95% CI:  $0.7, 3.2$ ). For BMI z-score, the direct effect estimates per  $10 \mu g/m^3$  was  $-0.4$  (95% CI:  $-1.5, 0.4$ ), and the indirect effect was  $0.1$  (95% CI:  $0.0, 0.3$ ), indicating a significant mediating role of inhibitory control. For percent body fat, the direct effect estimates per  $10 \mu g/m^3$  was  $-0.7\%$  (95% CI:  $-7.5, 5.0$ ), and the indirect effect estimate was  $3.0\%$  (95% CI:  $1.3, 5.0$ ). Lastly, for BMI change between ages 4 to 8 years, the direct effect estimates per  $10 \mu g/m^3$  was  $-0.6 \text{ kg/m}^2$  (95% CI:  $-2.6, 1.1$ ), and the indirect effect estimate was  $0.4 \text{ kg/m}^2$  (95% CI:  $0.1, 0.8$ ). The significant indirect effect estimates for all outcomes indicate that inhibitory control substantially and significantly mediated the relationship between postnatal  $PM_{2.5}$  exposure and childhood adiposity. These results suggest that behavioral changes linked to higher early-life air pollution

contribute to increased weight gain and growth trajectories. Supporting analyses showed that first-year postnatal PM<sub>2.5</sub> exposure was associated with inhibitory control class (omnibus  $p = 0.0098$ ), with marginal PM<sub>2.5</sub>  $\times$  inhibitory control class interactions for BMI change from 48 to 96 months ( $p = 0.074$ ) and BMI at 96 months ( $p = 0.093$ ), consistent with the exposure-mediator interaction structure used in the counterfactual mediation models.

## DISCUSSION

Our goal was to investigate whether higher early-life PM<sub>2.5</sub> exposure programs children's inhibitory control patterns, categorized by latent profiles, which then act as mediators of later-life child obesity. While prenatal PM<sub>2.5</sub> was not associated with obesity through inhibitory control, we found that first-year postnatal PM<sub>2.5</sub> exposure had indirect effects on multiple obesity-related outcomes via these neurobehavioral profiles. Previous research has shown that poorer inhibitory control is associated with obesity. Our findings provide clear and distinctive in demonstrating an indirect effect of upstream neurotoxic exposures, such as PM<sub>2.5</sub>, on obesity risk through a pathway involving inhibitory control as a mediator. Our study also shows that the definition of "obesogen" should be expanded to include neurotoxic effects. The neurotoxicity pathway for obesogens may not involve endocrine disruption and may instead be a neuroprogramming effect. Our results show that the effect of PM<sub>2.5</sub> exposure in the first postnatal year on obesity-related outcomes is mediated at least in part through changes in children's neurobehavior and likely arise from early-life PM<sub>2.5</sub>-induced neuro-inflammation. While this does not mean that the altered metabolism of food or dysregulation of hormones that regulate energy and appetite cannot play a role in the obesogenic properties of air pollution, it does suggest that at least a portion of the effect is likely due to altered eating behaviors related to inhibitory control that are programmed early in life. Our results also suggest that postnatal exposure to PM<sub>2.5</sub> during the first year of life is the window of vulnerability for this effect on inhibitory control capabilities, which in turn leads to increased childhood OWOB rates. This mediation highlights the importance of neurobehavioral pathways in understanding how environmental exposures influence child health beyond neurodevelopment and suggests that interventions aimed at modifying impulsive behavior could potentially mitigate the adverse effects of PM<sub>2.5</sub> on obesity in children.

Our findings align with previous research, demonstrating that poorer inhibitory control is associated with a higher risk of obesity. For example, poor inhibitory control has been shown to mediate the relationship between cortical thickness in prefrontal cortex (PFC) sub-regions and BMI in healthy adults [50]. Similarly, a cross-sectional study found that disgust sensitivity—an emotion that drives food avoidance—was associated with BMI, and this relationship was mediated by food-specific inhibitory control [51]. In essence, evidence highlights inhibitory control as a fundamental neurobehavioral correlate of BMI [52]. To our knowledge, we are the first study to report that patterns of inhibitory control mediate the relationship between environmental chemical exposures, specifically early-life PM<sub>2.5</sub> and childhood OWOB.

Identifying that inhibitory control serves as a mechanistic neurobehavioral pathway for the obesogenic properties of PM<sub>2.5</sub> is a previously underexplored and critical area of investigation as it points to behaviors as the causal pathway rather than metabolism. By understanding this pathway, we can better plan effective preventative interventions. Governmental and political commitment and coordination are necessary to implement more stringent, sustainable air quality policies. These policies should facilitate upstream interventions, such as transitioning to clean energy sources and strengthening industrial emission regulations, which represent the most impactful strategies for reducing community-level air pollution and mitigating related health risks, particularly in Mexico City. From a policy perspective, reducing air pollution would have substantial benefits in reducing its neurotoxicity and obesogenic properties together. Therapies for improving inhibitory control, such as cognitive behavioral therapy, may

also be more effective for weight loss than interventions designed to impact food metabolism. As previously mentioned, inhibitory control is a multiplex cognitive construct, enabling humans to voluntarily regulate their behavior by suppressing or resisting an impulsive response or temptation [53]. Structurally, a decentralized network of neuroanatomical regions (i.e., anterior cingulate, basal ganglia, cerebellum, and parietal cortex) supports inhibitory control, which undergoes maturation throughout childhood. The PFC is responsible for executing and modulating inhibitory control, with studies showing that children recruit activity in this region during inhibitory control tasks [54-56]. Studies have shown that PM<sub>2.5</sub> can cross the blood-brain barrier which is not fully developed in early childhood. This can negatively alter the structure and function of the PFC, possibly via PM<sub>2.5</sub>-mediated mechanisms that may play a role in inhibitory control [57, 58] by inducing neuroinflammation and oxidative stress [59, 60]. Early-life PM<sub>2.5</sub> has been linked to thinner cortical thickness in the PFC and maladaptive inhibitory control [61]. These prior findings support our results, suggesting that the neurotoxic dose-response effects of PM<sub>2.5</sub> may disrupt the neuroanatomical mechanisms underlying inhibitory control [62]. This impairment in inhibitory control could exacerbate maladaptive neurobehavioral patterns related to eating, thereby contributing to childhood OWOB.

Furthermore, the assessment of distinct inhibitory control profiles on the K-CPT in school-age children was conducted using an LPA methodology, a probabilistic, model-based variant of traditional nonhierarchical cluster analysis. The LPA was employed as a neurobehavioral phenotypic paradigm [63], in contrast with traditional statistical methods such as regression modeling that assume linearity and homogeneity and fail to recognize that neurobehavioral constructs like inhibitory control function are complex and may manifest differently across children. We found this approach beneficial in understanding how three different neurobehavioral profiles of inhibitory control relate to the link between PM<sub>2.5</sub> exposure and obesity, allowing for a more detailed interpretation of the data. Similarly, prior studies have utilized an LPA method to assess inhibitory control. For example, Martz et al. [64] used neuroimaging-based brain activation during inhibitory tasks, highlighting differences in neural activation patterns. Likewise, Malagoli et al. [65] identified inhibitory control profiles using survey-based inhibition assessments, distinguishing individuals with impulsive versus reflexive inhibitory control. While our study differs in methodology, our profiles share conceptual similarities, capturing variations in inhibitory function across individuals. We identified three inhibitory control profiles based on error rates and response speed variability in K-CPT tasks. Group 1 displayed a pattern of moderate inattentiveness and impulsivity with moderate variability in reaction times. Group 2 demonstrated the most consistent performance with fewer errors, indicating better attention and effective inhibitory control. In contrast, Group 3 exhibited a pattern of high impulsivity and the poorest overall performance, reflecting significant deficits in inhibitory control. Our findings parallel the limited existing data, suggesting distinct patterns of inhibitory control function. These results indicate that it is vital to identify neurobehavioral profiles that vary in inhibitory control efficiency, further emphasizing the heterogeneity in cognitive performance related to behavioral regulation.

Firstly, our investigation has several strengths, including a prospective study design in an urban population with historically higher exposure to air pollution on average and high obesity rates. Our longitudinal design included multiple rigorously collected obesity related measurements (BMI, BMI z-scores, and percent body fat) at two time points allowing for analysis of obesity changes over time. Our prospective design allowed for the investigation of inhibitory control as a neurobehavioral mediator between PM<sub>2.5</sub> and childhood OWOB as it was measured temporally at age 4-5 years before the childhood OWOB measures and years after the PM<sub>2.5</sub> critical window of infancy. Additionally, our statistical approach allowed us to identify latent profiles of inhibitory control based on the underlying data structure instead of imposing predefined categories. This approach also enabled us to integrate multiple subtasks of the K-CPT, providing an advantage over methods that analyze tasks in isolation and may overlook critical interdependencies. Some limitations of this study include a relatively small sample size and limited generalizability to other child populations, as our cohort resides in the metropolitan area of

Mexico City. The observed indirect effects represent the overall mediating influence of inhibitory control across all latent profiles. However, we acknowledge as a limitation that our analysis did not differentiate the specific profiles contributing to these effects, which could offer deeper interpretive insight into the observed associations. In addition, selection bias cannot be entirely ruled out and remains a possible limitation, as our study population experienced participant loss to follow-up over time. Participants who remained in the study population may differ systematically from those who were ultimately lost to follow-up, which could influence the generalizability of our results. Loss to follow-up was slightly associated with lower socioeconomic status, which may introduce selection bias if SES is related to both exposure and outcome. This could potentially bias the observed associations and should be considered when interpreting the findings. We further acknowledge that live birth bias may influence the observed associations if prenatal PM<sub>2.5</sub> exposure also affects the likelihood of non-live birth. Thus, the analytic sample may underrepresent more highly exposed or susceptible pregnancies, potentially biasing associations toward the null and underestimating true effects. Lastly, we acknowledge the potential influence of unaccounted covariates such as caloric intake and physical activity behaviors and the inability to assess detailed information on housing characteristics, such as cooking practices and ventilation systems, which would have allowed us to account for potential indoor sources of PM<sub>2.5</sub>. Future research should incorporate neuroimaging techniques, such as magnetic resonance imaging (MRI), to complement behavioral measures of inhibitory control by identifying the brain regions and neural pathways affected by PM<sub>2.5</sub> exposure. While inhibitory control is inherently a behavioral construct, understanding the neural substrates that support these behaviors can uncover their connection to obesity outcomes. This approach could provide deeper insights into the neurobehavioral mechanisms underlying PM<sub>2.5</sub>-induced obesity.

## Conclusions

Our study shows that greater early exposure to PM<sub>2.5</sub> in the first year of life is associated with alterations in inhibitory control function in childhood. These inhibitory control behaviors, in turn, predispose children to obesity, likely through increased impulsive eating behaviors. Based on existing evidence, this study provides new insights into the underlying mechanisms of the public health challenges to preventing obesity and promoting healthy child development. Our findings emphasize the necessity of ongoing initiatives to inform health and environmental policy and introduce behavioral (e.g., use air purifiers, monitor air quality, and encourage protective masks), and environmental measures to reduce air pollution and protect the health and well-being of children.

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### Author contributions

**JML**: Conceptualization, Methodology, Data interpretation, Writing—original draft, Writing—review & editing, Visualization. **HHLH**: Conceptualization, Formal analysis, Methodology, Data interpretation, Visualization, Writing—original draft, Writing—review & editing. **JML** and **HHLH** contributed equally to this work. **EOV**: Project administration, Data curation. **IK**, **EO**, **ACJ**, **VM**, **IA**, and **DNK**: Writing – review & editing. **BAC**: Data interpretation, Writing – review & editing. **MMTR**: Investigation, Resources, Project administration, Data curation, Writing – review & editing. **ROW**: Conceptualization, Investigation, Resources, Supervision, Project administration, Funding acquisition, Writing – review & editing, review of final draft.

### Competing Interests

The authors declare no competing interests.

### Ethics and approval and consent to participate

The study was approved by the institutional review boards of the Icahn School of Medicine at Mount Sinai [IRB protocol number: 12-00751A], and all three committees [Research, Ethics in Research, and Biosafety] were included in the IRB at the National Institute of Public Health in Cuernavaca, Mexico. Informed written consent was obtained from all maternal parents during the first visit, and all methods were performed in accordance with the relevant guidelines and regulations.

### Data Availability

All data used in the study's statistical analyses are publicly available through the Zenodo data and code repository [66; <https://zenodo.org/records/15692815>]. The available dataset includes demographic characteristics, PM<sub>2.5</sub> concentrations, and the neurobehavioral and BMI outcomes presented in Table 1. Source data for Figure 1 are provided in Zenodo data repository [66]. Source data for Figure 2 are provided in [Supplementary Data](#). All other data are available upon request to [robert.wright@mssm.edu](mailto:robert.wright@mssm.edu).

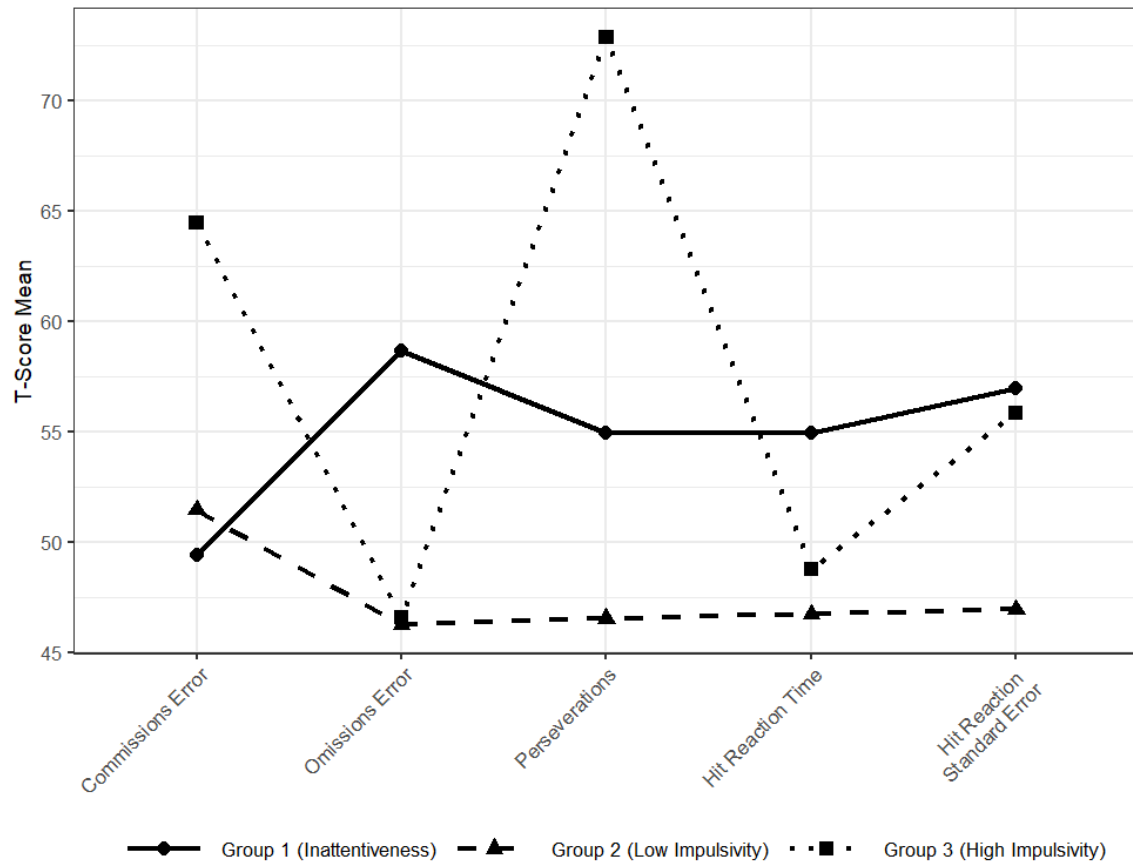
### Code Availability

Original code available in Zenodo Data [66]: <https://zenodo.org/records/15692832>

**Table 1**Overall demographic characteristics, PM<sub>2.5</sub> concentrations, neurobehavioral and BMI outcomes (N = 434)

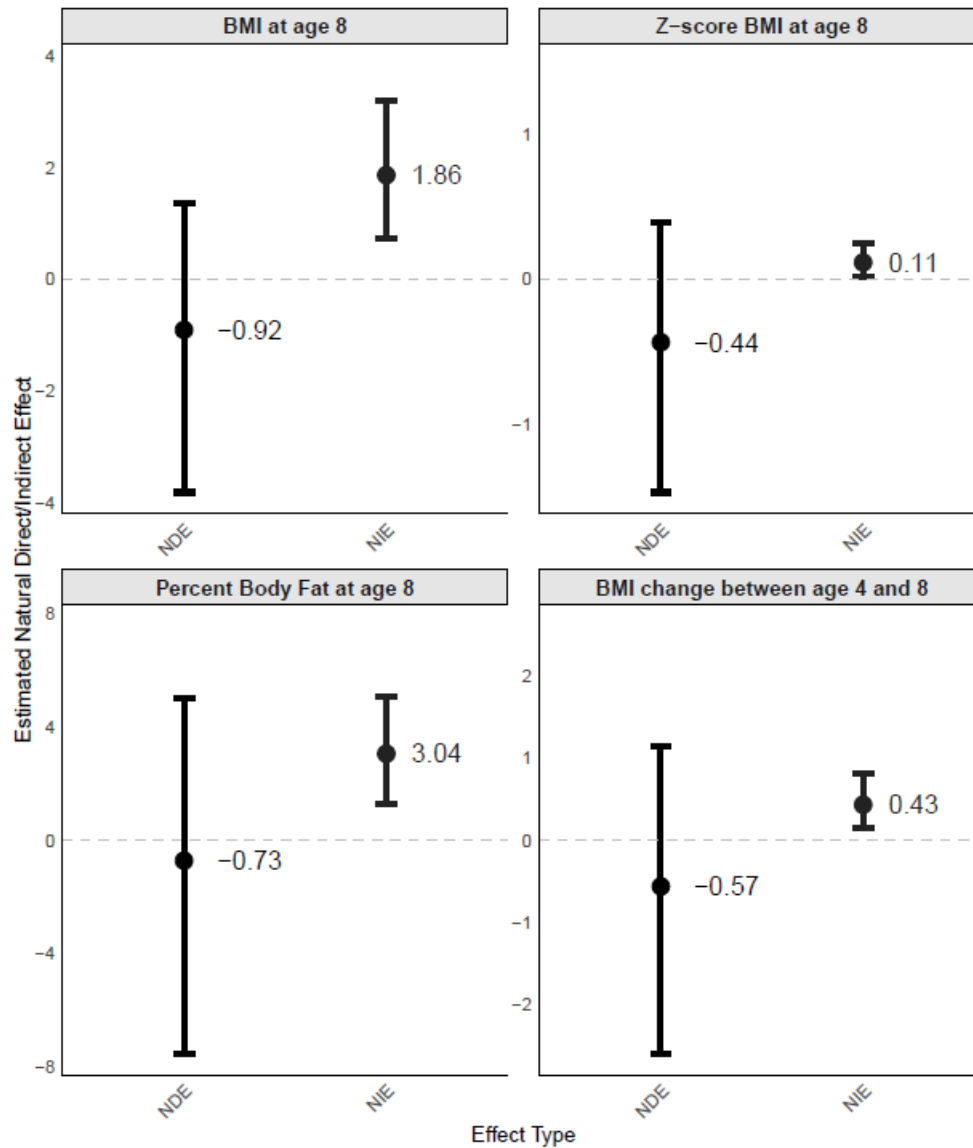
|                                                               | Overall            |
|---------------------------------------------------------------|--------------------|
|                                                               | Mean $\pm$ SD or % |
| Child sex                                                     |                    |
| Male                                                          | 219 (50.5%)        |
| Female                                                        | 215 (49.5%)        |
| Child age (years)                                             | 9.7 $\pm$ 0.7      |
| Mother age at delivery (years)                                | 28.21 $\pm$ 5.8    |
| Maternal SES                                                  |                    |
| Low                                                           | 228 (52.3)         |
| Medium                                                        | 162 (37.5)         |
| High                                                          | 44 (10.2)          |
| Neurobehavioral measures                                      |                    |
| Commissions T-scores                                          | 56.7 $\pm$ 9.7     |
| Omissions T-scores                                            | 48.6 $\pm$ 8.3     |
| Perseverations T-scores                                       | 59.3 $\pm$ 18.7    |
| Hit RT T-scores                                               | 49.4 $\pm$ 9.3     |
| Hit RT Standard Error T-Scores                                | 52.6 $\pm$ 7.5     |
| PM <sub>2.5</sub> concentrations ( $\mu\text{g}/\text{m}^3$ ) |                    |
| Prenatal PM <sub>2.5</sub>                                    | 22.7 $\pm$ 2.6     |
| Postnatal PM <sub>2.5</sub>                                   | 22.7 $\pm$ 2.5     |
| BMI measures                                                  |                    |
| BMI ( $\text{kg}/\text{m}^2$ ) at age 8                       | 18.8 $\pm$ 3.6     |
| BMI z-scores at age 8                                         | 0.8 $\pm$ 1.3      |
| PBF (%) at age 8                                              | 30.3 $\pm$ 8.8     |
| BMI change between ages 4 to 8 ( $\text{kg}/\text{m}^2$ )     | 3.1 $\pm$ 2.7      |
| Inhibitory control profiles                                   |                    |
| Inattentiveness                                               | 70 (16.1)          |
| Low impulsivity                                               | 178 (41.0)         |
| High impulsivity                                              | 186 (42.9)         |

Note: SD = Standard Deviation; Hit RT = Hit Reaction Time; PM<sub>2.5</sub> = Particulate Matter 2.5; BMI = Body Mass Index; PBF = Percent Body Fat.

**Figure 1:** Latent Profile Analysis of Inhibitory Control Patterns across K-CPT Indicators

Latent profile plot illustrating K-CPT performance, where the x-axis displays the five indicator variables used in the latent profile analysis (LPA) and the y-axis represents the mean scores. The three lines depict distinct inhibitory control patterns corresponding to the three identified neurobehavioral profiles. Analyses included  $n = 434$  children with complete data on K-CPT indicators.

**Figure 2:** Natural Direct and Indirect Effects of PM<sub>2.5</sub> Exposure on Obesity-Related Outcomes Mediated by Inhibitory Control Profiles



Estimated natural direct effects (NDE) and natural indirect effects (NIE) of first-year PM<sub>2.5</sub> exposure (per 10  $\mu\text{g}/\text{m}^3$  increase) on childhood adiposity outcomes. Points represent model-based effect estimates, and error bars represent 95% bootstrap confidence intervals from 2,000 resamples. Analyses included  $n = 434$  children with complete data on exposure, mediator, covariates, and outcomes. Since this mediation analysis was conducted in a single analytic sample rather than across groups, the same  $n$  applies to all estimates.

**ED SUM:**

Lane, Hsu et al. model ambient PM<sub>2.5</sub> exposure during pregnancy and the first year of life to examine how perinatal PM<sub>2.5</sub> contributes to childhood obesity via a pathway mediated by PM<sub>2.5</sub>-induced alterations in inhibitory control. They find that PM<sub>2.5</sub> exposure in infancy indirectly predicts obesity through its impact on inhibitory control deficits.

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